```
root@dansayag:~/pp/209877414/targil1/q3# gcc -m32 -g -o func func.c
root@dansayag:~/pp/209877414/targil1/q3# ls
a.out  func  func.c
```

3.1

```
root@dansayag:~/pp/209877414/targil1/q3# gdb func
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
    <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from func...
(gdb) break func2
Breakpoint 1 at 0x11ae: file func.c, line 5.
(gdb) where
No stack.
(gdb) run
Starting program: /root/pp/209877414/targil1/q3/func
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, func2 (a=269, b=6246) at func.c:5
```

3.2

```
(gdb) info registers ebp
ebp            0xffffd2b8          0xffffd2b8
```

3.3

ebp: מצביע לתחילת ה- Stack frame של הפונקציה הנוכחית.

$ebp = \underbrace{ffff\ d2b8}_{ffffd2c0} + 8 = a$

3.4

$a + 4 = b = ffffd2c4$

$a = ffffd2c0$

$b = ffffd2c4$

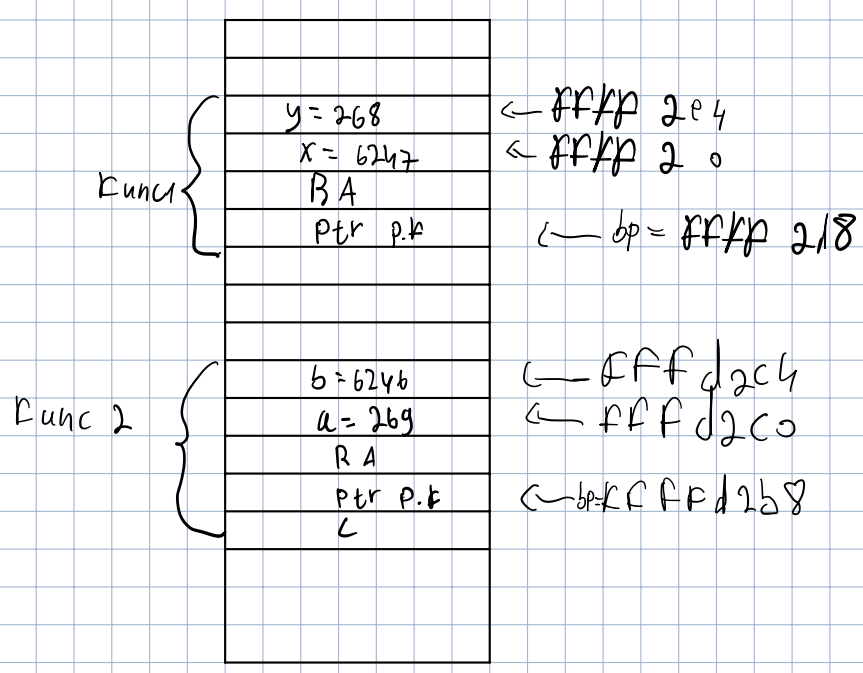
3.5

```
Breakpoint 1, func2 (a=269, b=6246) at func.c:5
5           c = a + b ;
(gdb) where
#0  func2 (a=269, b=6246) at func.c:5
#1  0x565561fc in func1 (x=268, y=6247) at func.c:10
#2  0x5655622f in main () at func.c:14
(gdb) info registers ebp
ebp            0xffffd2b8          0xffffd2b8
(gdb) x /wx $ebp+8
0xffffd2c0:     0x0000010d
(gdb) x /wx $ebp+12
0xffffd2c4:     0x00001866
(gdb) print &a
$1 = (int *) 0xffffd2c0
(gdb) print &b
$2 = (int *) 0xffffd2c4
(gdb)
```

3.6

```
0xffffd2c4:     0x00001866
(gdb) print &a
$1 = (int *) 0xffffd2c0
(gdb) print &b
$2 = (int *) 0xffffd2c4
(gdb) x /d $ebp+8
0xffffd2c0:     269
(gdb) x /d $ebp+12
0xffffd2c4:     6246
```

3.7

```
(gdb) print *((int*)$ebp+2)
$3 = 269
(gdb) print *((int*)$ebp+3)
$4 = 6246
(gdb) backtrace
#0  func2 (a=269, b=6246) at func.c:5
#1  0x565561fc in func1 (x=268, y=6247) at func.c:10
#2  0x5655622f in main () at func.c:14
(gdb)
```

3.8